Preparing your simulation Environment:

To setup your environment you must enter few commands in Linux terminal. Once you open a new terminal the prompt will be flashing at the end of a line that looks like this one:

```
name@ubuntu:~$|
```

Depending on your Linux distribution you might need to enter the following commands to invoke the `tcsh`or`csh`shell, and set CAD_HOME environment variable, respectively:

```
name@ubuntu:~$tcsh
name@ubuntu:~$ setenv  CAD_HOME /your-dir/magic
```

Then to change the directory to wherever you install Magic, use:

```
name@ubuntu:~$ cd /your-dir/magic/bin
```

Some other Linux commands that you may need:

Command	Description
`pwd`	(**P**rint **W**orking **D**irectory) that shows the current directory.
`ls`	**l**i**s**t the files and folders in the current directory.
`cd[directory-name]`	**C**hange **D**irectory to `[directory-name]`.
`cd ..`	Step out of the current directory to the parent directory.
`sudo [command]`	Runs `[command]`in administrator (super user) mode.

Build your design using Magic:

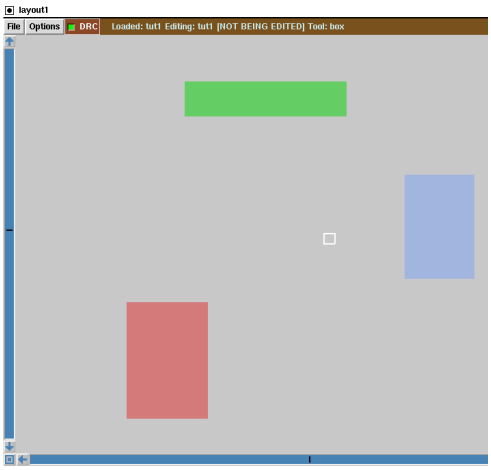
To start Magic, use this command:

```
name@ubuntu:~$ ./magic [your-design]
```

Assuming you want to build a Not gate, change "`[your-design]`" to "Not". This is also true for all other commands that contains wording between brackets (i.e`[your-design])`. The command will look like:

```
name@ubuntu:~$ ./magic Not
```

Once you press enter, Magic will run. It has two main windows. One for designing your layout, Figure 1, and the other for entering commands, namely tkcon2.3 Main which is shown in Figure 2.



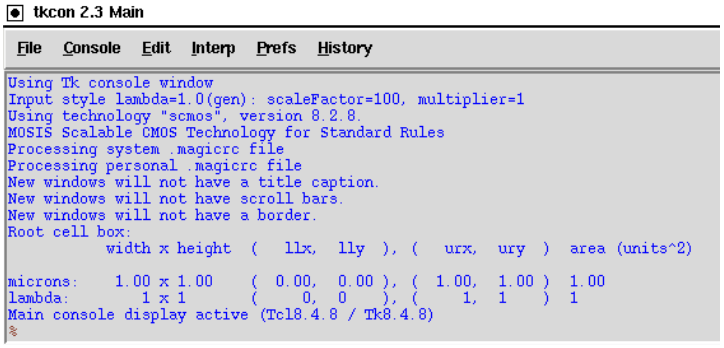


Figure 1: layout windowFigure 2: tkcon 2.3 main window

Build your design or gate by entering commands in tkcon window. Refer to Magic documentation to see list of all commands at http://www.opencircuitdesign.com/magic/index.html. The most frequent command in your project are:

Command	Description
`grid 4`	Grid the layout so that everbox is 4x4
`paint poly`	Paint polysilicon in red
`paint ndiff`	Paint n-diffusion in green
`paint pdiff`	Paint p-diffusion in orange
`paint m1`	Paint metal 1 in blue usually vertically
`paint m2`	Paint metal 2 in purple usually horizontally
`paint ndc`	Paint n-diffusion contact between ndiff and m1
`paint pdc`	Paint p-diffusion contact between pdiff and m1
`paint pc`	Paint polysilicon contact between poly and m1
`paint via`	Paint via contact between m1 and m2
`box`	Return the height and width of the selected area
`getcell [name of cell]`	Import old design in your current design. For example, `getcell Not`
`expand`	Display the selected, imported design
`unexpand`	Hide the expanded design
`upsidedown`	Flip your design around x-axis
`label [label-name]`	Move `[label-name]` to the current curser position

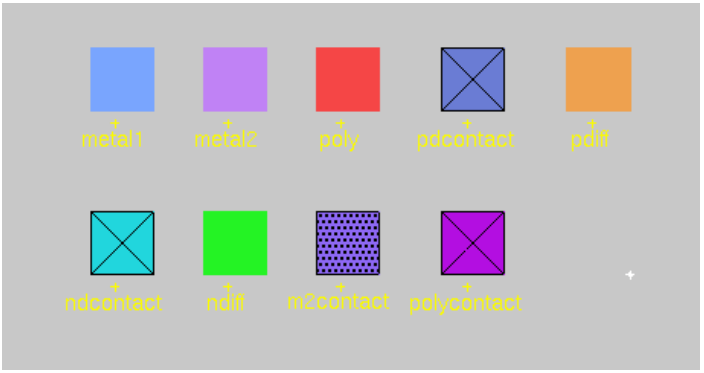


Figure 3: Different polygons in Magic

Basically, the most frequent command in your design is going to be`paint`. Figure 3 shows different polygons painted using Magic.
In addition to the above commands, there are several Magic controls which facilitate your work (but make sure the caps lock is off first):

- Press Z to zoom in and Shift+Z to zoom out.
- Ctrl + I: select all
- Right click then left click to create
  - a box or
  - a “+” for labeling.
- Press A to select the area, press Shift+A to unselect other areas
- Press D to delete the selected area.

- To lengthen or shorten an area:
  a. Select it
  b. Use the numbers:

1	2	3	4	5	6	7	8	9	0

  c. Or keypad:

7	8	9
4	5	6
1	2	3

To progress in your design, keep painting different layer of your layout as you design it in your stick diagram and make sure you always satisfy the design rules. If you violate any design rule between any polygons you will see white dots in your layout, Figure 4.

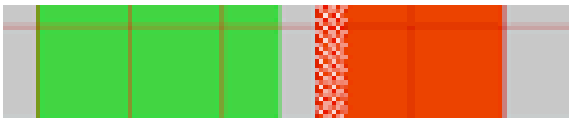


Figure 4: Violating design rule between n-diffusion and poly

After you finish building your design, label vdd, gnd, and all inputs and outputs by using the command:

```
label [label-name]
```

The label-name is going to be used in the simulation using Irsim later.When your done designing and labeling enter the following commands in order, to save your design and extract it:

```
save [your-design]
extract all
ext2sim [your-design].ext
q
```

Your design extracted file `[your-design].ext` will be converted to simulation file `[your-design].sim` by using `ext2sim [your-design].ext`

Simulate and test using Irsim:

Irsim is a tool for simulating digital circuits and is developed in Stanford.It is strongly associated with your Magic design, even though Magic is developed in Berkeley. To simulate using Irsim use the command

```
name@ubuntu:~$ ./irsim scmos100.prm [your-
                design].sim -[your-design].cmd
```

where `[yourdesign].sim` will be replaced by `Not.sim` in our case and the command file `[your-design].cmd` will be replaced by `Not.cmd`.

This will start Irsim which is composed of 2 windows. One to enter your Irsim commands depicted in Figure 5. And another window to show your simulated results and waveforms, Figure 6.

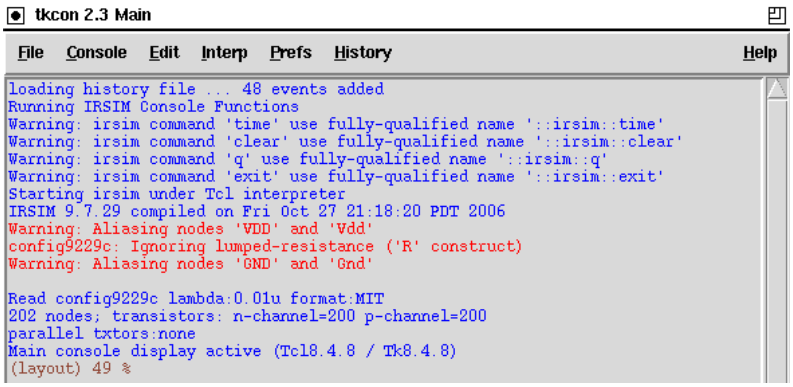


Figure 5: The Irsim command console

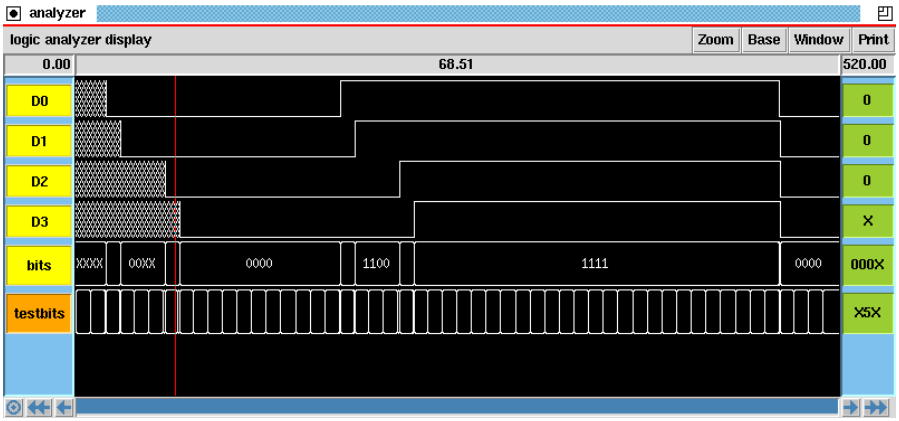


Figure 6: The "analyzer" graphic display in Irsim

The command file contain set of Irsim commands. Assuming your labels are: vdd, gnd, in and out, an example of command file, `Not.cmd` will look like:

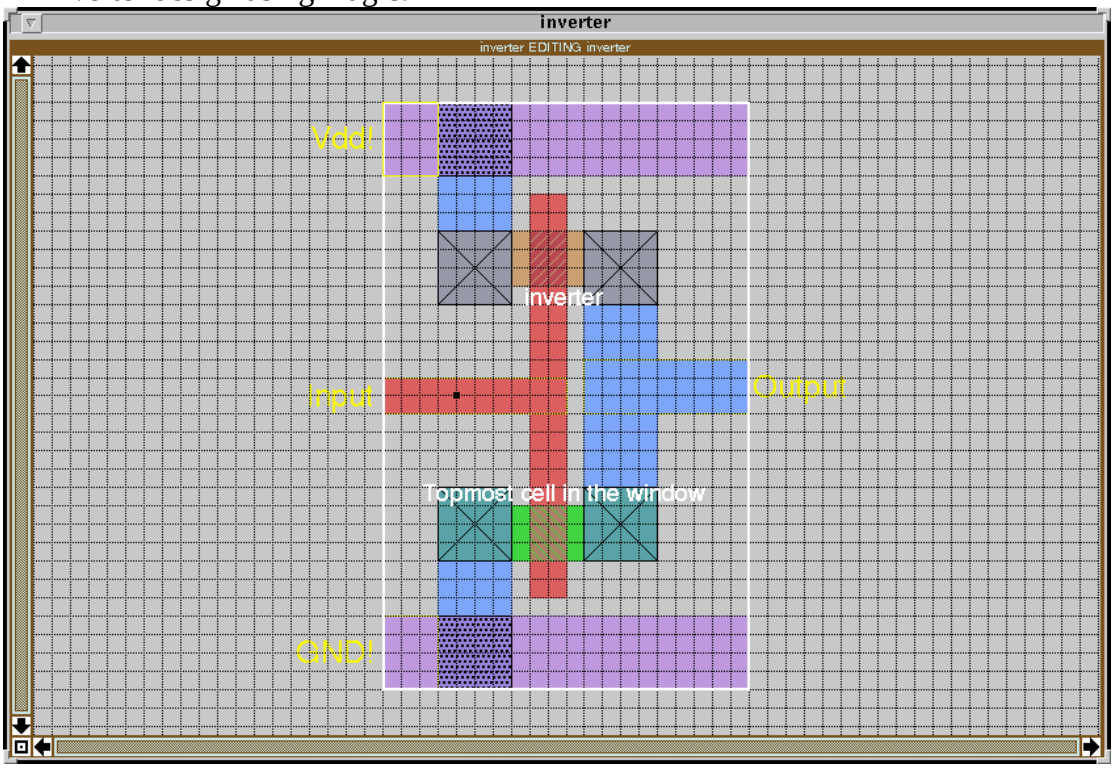
```
Stepsize 50
Ana in out
w in out
h vdd
l gnd
l in
s
h in
s
```

There is a long list of commands that you can use in Irsim. They can be found at http://opencircuitdesign.com/irsim. However, the most common Irsimcommands that you are going to use in your project are:

Command	Description
`Stepsize[integer]`	At every step move 50ns
`analabel1 label2 …`	Display the labels in the (**Ana**lyzer) window.
`w label1 label2 …`	**W**atch all the labels you give it.
`d`	**D**isplay the values at the labels you chose to watch (*only do this once*).
`s`	**S**tep the simulation (*i.e. advance time*).
`l label1 label2 …`	Set the labels to (**L**ow).
`h label1 label2 …`	Set the labels to (**H**igh).

You can use the command window of Irsim to enter the command directly instead of using a command file too. However, this means the commands will be lost once you close Irsim.

Example:
An inverter design using magic:



Exercise: Simulate your own inverter design using Irsim.